



Madhubala Priya

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ABOUT ME

Professional Summary

Passionate GIS and Remote Sensing professional with expertise in geospatial technologies, spatial data analysis, and web-based GIS applications. Skilled in Geographic Information Systems (GIS), Remote Sensing, spatial modeling, and geospatial data visualization, with hands-on experience in flood risk assessment, land use and land cover (LULC) analysis, disaster management, and environmental studies. Proficient in developing interactive GIS applications and applying geospatial technologies to support data-driven decision-making in urban planning, environmental management, and sustainable development.

Core Competencies

- Geographic Information Systems (GIS): Spatial analysis, mapping, geospatial data management, and spatial database development.
- Remote Sensing & Earth Observation: Satellite image processing, LULC analysis, change detection, and environmental monitoring.
- GIS Programming & Automation: Python, SQL, Google Earth Engine (GEE), and geospatial workflow automation.
- Machine Learning & Spatial Analytics: Supervised and unsupervised learning, predictive modeling, spatial pattern analysis, and AI-driven geospatial applications.
- Web GIS Development: Interactive mapping applications using HTML, CSS, JavaScript, and GIS web technologies.
- Spatial Data Visualization: Cartographic design, interactive dashboards, and data storytelling.
- Geospatial Software: QGIS, ArcGIS Pro, ERDAS Imagine, GeoDa, and geospatial data processing tools.
- Research & Analytical Skills: Spatial modeling, flood risk assessment, disaster management, environmental analysis, and data-driven decision-making.

Key Achievements

- Conducted geospatial research on flood risk assessment and disaster vulnerability mapping using GIS and Remote Sensing techniques.
- Completed thesis titled **"Flood Risk Assessment of Khagaria District, Bihar Using Geospatial Techniques"**, integrating GIS, Remote Sensing, and spatial modeling to identify flood-prone areas and support disaster risk reduction strategies.
- Developed interactive GIS applications and portfolio projects for spatial data visualization and analysis.
- Applied geospatial technologies to analyze environmental and urban challenges and support informed decision-making.
- Demonstrated proficiency in geospatial databases, cartography, spatial analytics, and data-driven problem-solving for real-world applications.
- Strong understanding of geospatial databases, cartography, and spatial analytics for real-world applications.
- Published research works in Scopus-indexed Journals.

Areas of Interest

Disaster Management | Climate Change & Environmental Studies | Spatial Data Analytics | Web GIS Development | Sustainable Development |

What Sets Me Apart?

I combine strong analytical skills with practical expertise in GIS, Remote Sensing, and spatial technologies to transform geospatial data into meaningful insights. My approach emphasizes continuous learning, problem-solving, and the development of innovative geospatial solutions that contribute to sustainable development and informed decision-making.

WORK EXPERIENCE

 **Savitribai Phule Pune University – Pune, India**

Researcher

[20/04/2022 – Current]

Major Research Contributions

- Published research on flood susceptibility assessment in Khagaria District, Bihar, using Remote Sensing and GIS techniques.
- Conducted flood frequency analysis of the Bagmati River Basin using Gumbel's Method and Log Pearson Type III Distribution.
- Investigated the impact of floods on Land Use and Land Cover (LULC) transformation through geospatial analysis.
- Developed GIS-based flood vulnerability mapping using the Analytical Hierarchy Process (AHP).
- Assessed spatio-temporal vegetation dynamics using MODIS time-series vegetation indices.
- Analyzed socio-economic flood damages across Bihar using GIS and the Wroclaw Taxonomic Method.
- Developed geospatial portfolio projects and interactive GIS applications for spatial visualization and analytics.

Technologies & Tools

GIS | Remote Sensing | Machine Learning | Python | SQL | Google Earth Engine (GEE) | QGIS | ArcGIS Pro | ERDAS Imagine | GeoDa | SPSS | Spatial Statistics | Cartography | Web GIS

EDUCATION AND TRAINING

PhD in Geography

Savitribai Phule Pune University, Pune [20/04/2022 – Current]

City: Pune | Country: India

M.A in Geography

Banaras Hindu University, Varanasi [09/08/2018 – 17/11/2020]

City: Varanasi | Country: India | Field(s) of study: Earth sciences

B.A in Geography

Banaras Hindu University, Varanasi [20/08/2015 – 20/06/2018]

City: Varanasi | Country: India | Field(s) of study: Earth Sciences

LANGUAGE SKILLS

Mother tongue(s): Hindi

Other language(s):

English

LISTENING C2 **READING** C2 **WRITING** C2

SPOKEN PRODUCTION C2 **SPOKEN INTERACTION** C2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

SQL / Python / ArcGIS / QGIS / R / ENVI / Google Earth Engine / Git / JavaScript / Erdas Imagine / Geoserver / Machine Learning / Google Workspace (Google Drive, Google Docs, Google Slides, Google Sheets, Google Forms)

PUBLICATIONS

[2025]

Relief Analysis for Identification of Flood Susceptibility in Khagaria District, Bihar Using Remote Sensing and GIS Techniques Relief analysis describes an area's relief and landform structure. It is a vital source of information for assessing flood susceptibility in a region. Khagaria is one of the most flood-prone areas in Bihar, where floods significantly affect residents, the environment, and cause economic losses. This study examines seven topographic parameters, namely elevation, slope, hillshade, drainage density, TWI (Topographic Wetness Index), ruggedness number, and relative relief in Khagaria district. These factors provide deeper insights into the flood characteristics of the study area and offer observations on the structural landscape patterns. The relief-related maps have been prepared using satellite data and Digital Elevation Models (DEMs), employing Remote Sensing (RS) and Geographic Information System (GIS) techniques. According to the slope map, the terrain is mostly level, with an average elevation of 36 meters and a slight slope. Due to extensive lowland areas in the northern, north-eastern, and southern regions, along with the general relief characteristics, these areas are more prone to flooding. Write here the description...

Authors: Madhubala Priya 1* , Nitin Mundhe 2 , Avinash Shelar 3 | **Journal Name:** Iraqi National Journal of Earth Science | **Volume, Issue and Pages:** 25(4)

Priya, M., Mundhe, N., & Shelar, A. (2025). Relief analysis for identification of flood susceptibility in Khagaria District, Bihar using remote sensing and GIS techniques. INJES, 25(4), 273–290. <https://doi.org/10.33899/earth.2025.154846.1384>

Link: <https://iasj.rdd.edu.iq/journals/journal/view/472>

[2026]

Flood Frequency Analysis Along Bagmati River Basin in Khagaria District, Bihar Using Gumbel's Method and Log Pearson Type III Distribution Flood frequency analysis is an essential hydrological method for determining the size and recurrence intervals of flood events, especially in areas vulnerable to flooding. One such area where frequent flooding has a major effect on the environment, the economy, and local inhabitants is Khagaria in Bihar, India. This study uses the Log-Pearson Type III Distribution and the Gumbel Extreme Value Distribution to analyze historical discharge data spanning 40 years (1985-2024) from the Bagmati River Basin in Khagaria District. Important statistical measures, including the mean, standard deviation, and skewness, are used to predict catastrophic hydrological events and to assess flood magnitudes for various return periods. The findings show that while both methods effectively model catastrophic floods, the Log-Pearson Type III Distribution more accurately captures variability at longer return periods. The results indicate the highest recorded flood discharge of 30734.001 m³/s in 2023 and the lowest discharge of 884.77 m³/s in 2011. The analysis calculates the anticipated inundation for the return periods of 2, 10, 25, 50, 100, and 1000 years. The expected flood for the two-year return period is 2833.731 cumecs according to the Gumbel distribution, while the Log-Pearson III distribution predicts a flood of 1169.9 cumecs. It has been noted that Gumbel estimated higher values for all of the aforementioned return periods, except for 1000 years, where Log Pearson III predicted significantly higher values. The 2-year flood event has a 50% possibility of occurrence in any year with average impacts, while severe flooding events are predicted at longer return periods, with discharge values exceeding the river's carrying capacity. The results highlight the urgent need for proactive floodplain management, infrastructure planning, and risk reduction strategies by illustrating the region's increasing flood frequency and severity. This study provides valuable information for sustainable flood management, hydrological modelling, and infrastructure design in the area under study.

Authors: Priya Madhubala1*, Mundhe Nitin1 and Magar Arun2 | **Journal Name:** EARTH SCIENCES RESEARCH JOURNAL | **Volume, Issue and Pages:** 29(4)

Madhubala, P., Nitin, M., & Arun, M. (2026). Flood Frequency Analysis Along Bagmati River Basin in Khagaria District, Bihar Using Gumbel's Method and Log Pearson Type III Distribution. ESRJ, 29(4), 459-468. <https://doi.org/10.15446/esrj.v29n4.119636>

Link: <https://revistas.unal.edu.co/index.php/esrj/article/view/119636>

[2026]

Integrated Flood Hazard and Settlement Vulnerability Assessment in Khagaria District, Bihar Using Geospatial Techniques and the AHP Method The flood history of Khagaria district in Bihar has been catastrophic, with over 74% of the region susceptible to flooding. The significant floods that transpired in 1954, 1957, 1974, 1987, 2004, 2007, 2017, 2020, 2022, and 2024 profoundly affected individuals' lives, properties, and the district's agricultural sector. The current research seeks to employ a multicriteria decision-making framework within a Geographical Information System (GIS), integrating remotely sensed data, to delineate flood-prone areas in Khagaria district. The research employed an 11x11 decision matrix to assess the significance of each parameter in generating the final flood map via the Analytic Hierarchy Process (AHP). The parameters included are elevation,

slope, rainfall, drainage density (DD), distance from river (DFR), topographic wetness index (TWI), stream power index (SPI), land use/land cover (LULC), normalised difference vegetation index (NDVI), lithology, and ruggedness index. The weights derived in the case are accepted, as evidenced by a consistency ratio (CR) of 0.0345 (less than 0.1). Khagaria district encompasses a total area of 1490.16 sq.km, of which 429.909 sq.km is designated as a high flood susceptible zone, and 399.502 sq.km as a very high flood vulnerable zone. The integrated 2024 flood inundation raster was utilized to calculate the impacted settlement areas by village (in hectares and percentage) and their inundation frequency, with corresponding maps prepared for the year 2024. The vulnerability map was validated by the Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) methodology. The analysis produced an AUC value of 0.819 for the integrated inundation raster and 0.945 for the flood susceptibility zonation map, demonstrating a high degree of accuracy and a robust correlation with identified flood-prone areas, signifying a substantial association between the satellite flood data and the assessed flood-prone locations. The outcomes can substantially improve local emergency preparedness efforts, infrastructure advancement, land-use planning, and flood risk mitigation strategies.

Authors: Madhubala Priya 1, Umesh Potekar 2 and Nitin Mundhe3 | **Journal Name:** Journal of Water Management Modelling

[2026]

Quantifying Flood Impacts on Land Use and Land Cover and Predicting Future Landscape Dynamics Using Multi-Temporal Remote Sensing Data and the CA-Markov Model: A Case Study of Khagaria District, Bihar, India *The spatiotemporal analysis of land use and land cover is carried out from 2015 to 2025 using remotely sensed data, including Landsat 8 (2015), Sentinel-2 (2020) and Sentinel-2 (2025) of Khagaria district, Bihar. The study used a random forest classifier to perform supervised classification on the LU/LC map, producing five classes: forest, water bodies, built-up cropland, and barren landscape. The overall accuracy for the LU/LC maps in 2015, 2020, and 2025 was 95%, 91%, and 95%, respectively. It was discovered that the flood was the primary cause of the LU/LC shift. In this area, flooding is a common occurrence. The sector most affected by floods was agriculture, which had a 3.87% decrease after the flood event even though it remained the most agriculturally dominated sector. The built-up area has grown by 5.19%, with some parts farther away and others near flood-prone areas. The Forest/Plantation has shrunk by 1.53%. Floods have resulted in a 0.99% percent decrease in water bodies and a 1.19% percent increase in barren land. The government should enact appropriate legislation, including mandatory insurance coverage for assets located in high-risk areas, to ensure compensation for losses caused by natural disasters. Furthermore, Land Use/Land Cover (LU/LC) analysis plays a critical role in identifying the underlying drivers of adverse land cover change, thereby supporting evidence-based policymaking and sustainable land management.*

Authors: Priya Madhubala 1*, Potekar Umesh P. 2 and Mundhe Nitin3 | **Journal Name:** Disaster Advances

[2026]

Assessment of Spatio-Temporal Vegetation Dynamics using Modis time series vegetation Indices – A case study of Himachal Pradesh This study examined the spatio-temporal dynamics of vegetation to identify and characterize vegetation stress zones in Himachal Pradesh, India, utilizing the Normalized Difference Vegetation Index (NDVI) and Vegetation Condition Index (VCI). The indices were obtained from time-series MODIS 16-day composite products (MOD13Q1) at a spatial resolution of 250 m for all months from 2015 to 2022. Monthly NDVI layers were aggregated to obtain the yearly mean NDVI, and a long-term NDVI baseline was established by synthesizing multi-year NDVI composites. The Vegetation Condition Index (VCI) was calculated utilizing both long-term and current Normalized Difference Vegetation Index (NDVI) measurements and subsequently categorized into four classifications: High, Moderate, Low, and No Vegetation. The spatial expanse of each class was measured in square kilometers and expressed as a percentage. Data processing and analysis utilized ArcGIS 10.4, ERDAS Imagine, Notepad++, and MS Excel. The variability of VCI was assessed year-wise by a comparison analysis utilizing bar graphs. The findings indicate significant interannual variability in vegetation condition, with 2015 showing the least area under Moderate and High VCI, signifying extensive vegetation stress, while 2021 demonstrated the greatest extent of these categories, indicating favorable vegetation conditions due to enhanced climatic and monsoonal patterns. The research illustrates the effectiveness of multi-temporal MODIS-based VCI analysis in assessing vegetation health and identifying environmental stress in mountainous environments

Authors: Priya Madhubala 1*, Potekar Umesh P. 2 and Mundhe Nitin3 | **Journal Name:** Transactions

CERTIFICATIONS

[IIRS (ISRO), 28/10/2024]

One Day Workshop on Space based inputs for Village level crop assessment

Mode of learning: Online

[IIRS (ISRO), 18/12/2024]

RS & GIS Applications in Natural Resource Management

Mode of learning: Online

[Savitribai Phule Pune University, Pune, 06/02/2026]

Assessment of Spatio-Temporal Vegetation Dynamics using Modis time series vegetation Indices – A case study of Himachal Pradesh

Mode of learning: Presential

[Savitribai Phule Pune University, Pune,, 30/01/2026]

Assessing Socio-economic Flood Damage Across the Districts of Bihar: A Taxonomic Analysis and Spatial Evaluation

Mode of learning: Presential

[Aligarh Muslim University, 29/09/2024]

Analyzing the impact of flood on LULC transformation in Khagaria District, Bihar Using Remote Sensing and GIS Techniques

Mode of learning: Presential

[Ahmednagar College, Ahmednagar, 19/01/2023]

Quality of life of Construction Workers in BHU Campus, Varanasi

Mode of learning: Presential

PROJECTS

Flood Risk Assessment for Residential Areas in Miami The project focuses on assessing flood risk for residential areas in Miami using geospatial techniques and multi-source datasets. By integrating river networks, elevation data, and school locations, the study first evaluates the existing flood vulnerability scenario. Spatial analysis methods such as buffering, DEM reclassification, and overlay analysis are applied to identify flood-prone zones, particularly in proximity to critical infrastructure like schools. Further, zonal statistics are used to quantify the level of risk in these areas. The project delivers data-driven insights and decision-support outputs that can help urban planners and local authorities enhance flood resilience and improve infrastructure planning in vulnerable neighborhoods.

Integrated Flood Hazard and Settlement Vulnerability Assessment in Khagaria District, Bihar Using GIS, Remote Sensing, AHP, and Web GIS This project focuses on identifying flood-prone areas in Khagaria District, Bihar, by integrating GIS and Remote Sensing techniques with the Analytic Hierarchy Process (AHP). Eleven parameters, including elevation, slope, rainfall, drainage density, distance from rivers, land use/land cover (LULC), NDVI, lithology, and topographic indices, were analyzed to generate flood vulnerability maps. The model was validated using ROC-AUC analysis, demonstrating reliable prediction accuracy. Additionally, an **interactive Web GIS dashboard** was developed using **HTML, CSS, and JavaScript** to visualize flood risk zones, settlement vulnerability, building-level exposure, and nearest flood distances through interactive maps and pop-ups. The Web GIS platform enables planners and decision-makers to explore flood information dynamically and supports disaster management, infrastructure planning, and flood mitigation strategies in vulnerable regions.

Flood Susceptibility Analysis using PyQGIS & QGIS Automation In this project, I developed a workflow integrating: Remote Sensing & GIS techniques Analytical Hierarchy Process (AHP) DEM, Slope, Rainfall, Land Use/Land Cover (LULC), and Distance from River parameters PyQGIS scripting for automation and reproducible geospatial analysis The workflow includes: Data preprocessing and validation Raster reprojection, clipping, and resampling Parameter reclassification and weight assignment Weighted overlay analysis Generation of Flood Susceptibility Index (FSI) map The final output classifies the study area into different flood susceptibility zones, helping identify areas with higher flood risk and supporting disaster management, land-use planning, and mitigation strategies. A major highlight of this project was creating custom QGIS tools using PyQGIS to reduce manual processing time and improve automation in flood mapping workflows. Tools & Technologies: QGIS PyQGIS

Remote SensingGIS AnalysisAHP MethodologySpatial AnalysisThis project strengthened my skills in geospatial automation, environmental modelling, and applying GIS for real-world disaster management challenges.

Land Surface Temperature (LST) Analysis of Patna City Using Remote Sensing & GIS In this project, I developed an interactive visualization of Land Surface Temperature (LST) patterns in Patna, Bihar using geospatial technologies. **Tools & Technologies Used:**• Remote Sensing• GIS & Spatial Analysis• Satellite Data Processing• Land Surface Temperature Extraction• Web GIS Development• HTML, CSS & JavaScript**Project Highlights:**Mapping spatial variation of surface temperatureUnderstanding urban heat patternsVisualizing temperature distribution through an interactive web interfaceExploring the relationship between urbanization and thermal environmentLand Surface Temperature analysis plays an important role in studying Urban Heat Island (UHI) effects, climate resilience, and sustainable urban planning. Recent studies in Patna have also highlighted increasing surface temperature associated with urban expansion and changes in land use/land cover.This project helped me strengthen my skills in geospatial analysis, satellite data interpretation, and Web GIS application development.I look forward to applying these skills in more projects related to climate change, disaster management, and environmental monitoring.

Link: <https://lnkd.in/dN3wN4Bz>

Landslide Susceptibility Analysis for Wayanad District This project develops a landslide susceptibility map using the Analytical Hierarchy Process (AHP) integrated with GIS techniques. Multiple landslide conditioning factors, including slope, elevation, aspect, rainfall, lithology, land use/land cover (LULC), distance from rivers, NDVI, STI, TWI, and AOI, are utilized to assess landslide vulnerability. **The workflow is automated through PyQGIS scripting**, which performs layer validation, reprojection, clipping, resampling, reclassification, and weighted overlay analysis. Automation minimizes manual effort, improves accuracy, and ensures reproducibility of the landslide mapping process. **The final output is an AHP-based landslide susceptibility map and an interactive WebGIS platform**, enabling visualization, analysis, and dissemination of landslide risk information to researchers, planners, and decision-makers for effective hazard management and mitigation planning. **Objective- To create Landslide susceptibility map using the Analytical Hierarchy Process (AHP) integrated tool creation with PyQGIS Script and WEB Mapping.**

Quantifying Flood Impacts on Land Use and Land Cover and Predicting Future Landscape Dynamics Using Multi-Temporal Remote Sensing Data and the CA-Markov Model: A Case Study of Khagaria District, Bihar, India This project analyzes the changes in land use and land cover patterns in Khagaria District, Bihar, using Remote Sensing and GIS techniques. Multi-temporal satellite data from Landsat and Sentinel were used to map pre-, during-, and post-flood LULC conditions for 2015, 2020, and 2025. The study identified significant changes, including a decline in agricultural land and an increase in built-up areas due to recurrent flooding. Furthermore, the **CA-Markov model** was applied to predict future land-use scenarios for 2035, providing valuable insights for sustainable land management, disaster risk reduction, and flood-resilient planning in the region.

HOBBIES AND INTERESTS

Blogging

Reading

Basketball Playing

Writing Blogs

CONFERENCES AND SEMINARS

[18/01/2023 – 19/01/2023] Ahmednagar, Maharashtra, India

International Conference on Recent Trends in Geography and Environment Study Jointly Organized by Department of Geography and Maharashtra Bhugolshastra Parishad, Pune B.P.H.E Society's Ahmednagar College, Ahmednagar. (18th and 19th January 2023).

[28/09/2024 – 30/09/2024] Aligarh, Uttar Pradesh, India

International Conference on Global Climate Change: Resilient Society and Sustainable Development organized by department of Geography, Aligarh Muslim University, Aligarh (28th –30th September 2024).

[22/02/2025 – 24/02/2025] Gaya, Bihar, India

XIXth DGSi INTERNATIONAL GEOGRAPHY CONFERENCE, THE DECCAN GEOGRAPHICAL SOCIETY OF INDIA, Focal Theme- Geospatial Technology for Sustainable Development: A Way Forward for Viksit Bharat Organized By Department of Geography, Central University of South Bihar, Gaya (Bihar), (February, 22-24, 2025).

[30/01/2026 – 31/01/2026] Pune, Maharashtra, India

INTERNATIONAL CONFERENCE ON "WATER, LAND AND LIFE: GEOGRAPHICAL PERSPECTIVES ON SUSTAINABILITY" organized by Sir Parashurambhau College, Pune. (30-31st January 2026).

[04/02/2026 – 07/02/2026] Pune, Maharashtra, India

National Conference on Geospatial Technologies and sustainable Landscape: Navigating Urban-Rural Transformation and Resource Challenges (4-7th February 2026).

RECOMMENDATIONS

Name: Dr. Nitin Mundhe | Research Guide

I have known him for several years. He is my research guide in PhD. I strongly recommend. He has a PhD and Post-doctoral degree in Geography.

Email: mundhenitin8@gmail.com | Phone number: (+91) 9420343203

Name: Nagendra Singh | Mentor

He is the founder of two geospatial companies- Harita Hive, MaptrixAI and my mentor as well. He has a M.Sc degree in Geography and has worked experience of 8 years in geospatial industries. I am a learner at Harita Hive beside my PhD.

Email: singhnagendrageotech@gmail.com | Phone number: (+91) 9286833533

DECLARATION

I hereby declare that all the information furnished above is true to the best of my knowledge.